Qasim Taha

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SKILLS

CAD: SolidWorks (DFMA, Drawings), AutoCAD, GD&T
Analysis: ANSYS FEA/CFD, MATLAB, Root Cause Analysis
Documentation: MS Excel, Technical Drawings, BOMs, SOPs
Collaboration: Vendor Liaison, Cross-Functional Teams, Leadership

CERTIFICATIONS

SOLIDWORKS CSWP | *Certified SOLIDWORKS Professional* Sept. 2026
Issued by Dassault Systèmes
CSWA (Nov. 2023); Associate certifications in SolidWorks Electrical, Sustainability, and Additive Manufacturing (Sept. 2026)

EDUCATION

Bachelor of Engineering, Mechanical Engineering Co-op | *University of Guelph* Sept. 2022 – Apr. 2027
Guelph, ON

WORK EXPERIENCE

Manufacturing Engineer Co-op | *Roadtrek Inc.* May 2026 – Present
Cambridge, ON

- Designed a **drill jig** for a propane regulator bracket that fixtures the part and locates holes, cutting prep time by **67%**.
- Rebuilt kitting checklists with part ID fields, eliminating waits on missing parts and cutting downstream station time by **60%**.
- Led **material selection** for a failing butyl sealing tape, benchmarking supplier specs on adhesion, heat, and water sealing.
- Traced an assembly defect through **RCA** and worked with the vendor on a prototype part that resolved it.
- Cross-checked **BOMs** from **1000+** engineering drawings, building an Excel tool to flag mismatched part numbers.

Manufacturing Engineer Co-op | *Camcor, Linamar Corporation* Jan. 2025 – Aug. 2025
Guelph, ON

- Owned design through release of **18** production tools in **SolidWorks** for automotive PTU and RDM driveline programs.
- Produced detailed drawings to **ASME Y14.5 GD&T** for gear cutting fixtures, robot pallets, and bearing pullers.
- Laid out machine cells and the **plant floor layout** in **AutoCAD** for a new assembly line launch and set station **cycle time** targets.
- Designed variable and go/no-go inspection gauges to verify part quality on the new line's launch parts.
- Designed a spring-loaded oiling tool rated for **300 uses/shift**, calculating spring rate and guide sizing; cut cycle time **93%**.
- Authored Safe Cell Entry procedures and standard work instructions for new line stations, training operators through launch.

PROJECTS

Mobile Strawberry Pre-Cooling Cart | *Engineering & Design III* Sept. 2025 – Dec. 2025
University of Guelph, Guelph, ON

- Designed a battery-electric strawberry pre-cooling cart in a team of 5, selecting from 7 concepts via weighted decision matrices.
- Sized an **R-290** refrigeration system (1,640 W load, COP 2.09) and 48 V $LiFePO_4$ battery for **5.7 h** runtime, **43%** above target.
- Modeled the assembly in **SolidWorks** and validated it with **ANSYS CFD**, confirming cooling from 30.5 to 1 °C in **2.17 h**.
- Built a 10-year cash flow model projecting a **2-year payback** on a **$6,515** build, a quarter the cost of commercial units.

Detachable Conveyor Belt Cart | *Camcor, Linamar Corporation* Jan. 2025 – Aug. 2025
Guelph, ON

- Designed a detachable cart to move parts between conveyors, replacing manual carrying and cutting handling time by **70%**.
- Reverse-engineered existing conveyors by measuring dimensions and interface points, so the cart fits varying setups.
- Built the frame from **T-slot extrusion** and off-the-shelf casters, with a drive shaft coupling for tool-free attach and detach.
- Detailed the cart in **SolidWorks** to **ASME Y14.5 GD&T**; it mated with existing conveyor lines with no rework.

Additional projects, including a pick-and-place gripper and more tooling and fixture designs, are at qtaha.com.

EXTRACURRICULARS

Product Designer | *Straw Co (Part-Time)* June 2025 – Present

- Design products in **SolidWorks** for a medical-accessories brand, including the Straw V2 loupe strap, from concept to production.

3D Printing Business | *Self-Employed* June 2025 – Present

- Design and sell parts in carbon-fiber nylon (PA6-CF, PA12-CF) and ABS, optimized for tight tolerances and strength.

President | *Pakistani Students Association* Sept. 2024 – Apr. 2025